PULLBACK TO THE TOTAL SPACE OF A VECTOR BUNDLE IS AN ISOMORPHISM ON CHOW GROUPS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a completion to a projective bundle with the vector bundle realised as the complement of the zero section divisor, and an expansion through the projective bundle formula together with a vanishing pushforward computation

Editorial note. The label lemma-vectorbundle names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Surjectivity of pullback for a flat morphism of relative dimension r with affine r-space fibres, applied to the vector bundle
- The projective bundle formula for the projective bundle of the finite locally free sheaf with a trivial summand adjoined, and the identification of the complement of the zero section with the vector bundle
- The exact sequence describing the kernel of restriction to an open subspace as the cycles supported on the complement, the relation between the invertible sheaf of an effective Cartier divisor and the tautological quotient, and the compatibility of cap with pushforward

Source reference. Original reference 02TX: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 4381–4381. Prerequisite pointer invoked at proof line(s) 4558; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B74: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 4017–4017. Prerequisite pointer invoked at proof line(s) 4530; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02TT: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3948–3948. Prerequisite pointer invoked at proof line(s) 4531; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 02SU: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2933–2933. Prerequisite pointer invoked at proof line(s) 4565; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02TR: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3889–3889. Prerequisite pointer invoked at proof line(s) 4564; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RX: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1951–1951. Prerequisite pointer invoked at proof line(s) 4556; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 4516; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0847: sources/stacks-chow/tags.txt, source lines 10312–10312. Prerequisite pointer invoked at proof line(s) 4544; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let X/B be good. Let $\mathcal{E}$ be a finite locally free sheaf of rank r on X. Let*

$$p : E = \underline{\mathrm{Spec}}(Sym^*(\mathcal{E})) \longrightarrow X$$

be the associated vector bundle over X. Then $p^* : \mathrm{CH}_k(X) \to \mathrm{CH}_{k+r}(E)$ is an isomorphism for all k.

Proof. (For the case of linebundles, see Lemma 0B74.) For surjectivity see Lemma 02TT. Let $(\pi : P \to X, \mathcal{O}_P(1))$ be the projective space bundle associated to the finite locally free sheaf $\mathcal{E} \oplus \mathcal{O}_X$. Let $s \in \Gamma(P, \mathcal{O}_P(1))$ correspond to the global section $(0, 1) \in \Gamma(X, \mathcal{E} \oplus \mathcal{O}_X)$. Let $D = Z(s) \subset P$. Note that $(\pi|_D : D \to X, \mathcal{O}_P(1)|_D)$ is the projective space bundle associated to $\mathcal{E}$. We denote $\pi_D = \pi|_D$ and $\mathcal{O}_D(1) = \mathcal{O}_P(1)|_D$. Moreover, D is an effective Cartier divisor on P. Hence $\mathcal{O}_P(D) = \mathcal{O}_P(1)$ (see Divisors on Spaces, Lemma 0847). Also there is an isomorphism $E \cong P \setminus D$. Denote $j : E \to P$ the corresponding open immersion. For injectivity we use that the kernel of

$$j^* : \mathrm{CH}_{k+r}(P) \longrightarrow \mathrm{CH}_{k+r}(E)$$

are the cycles supported in the effective Cartier divisor D, see Lemma 02RX. So if $p^*\alpha = 0$, then $\pi^*\alpha = i_*\beta$ for some $\beta \in \mathrm{CH}_{k+r}(D)$. By Lemma 02TX we may write

$$\beta = \pi_D^*\beta_0 + \ldots + c_1(\mathcal{O}_D(1))^{r-1} \cap \pi_D^*\beta_{r-1}.$$

for some $\beta_i \in \mathrm{CH}_{k+i}(X)$. By Lemmas 02TR and 02SU this implies

$$\pi^*\alpha = i_*\beta = c_1(\mathcal{O}_P(1)) \cap \pi^*\beta_0 + \ldots + c_1(\mathcal{O}_D(1))^r \cap \pi^*\beta_{r-1}.$$

Since the rank of $\mathcal{E} \oplus \mathcal{O}_X$ is $r + 1$ this contradicts Lemma 02SU unless all α and all β_i are zero. □